

What is System Modeling and UML?

What is System modelling?

System modelling is the process of developing abstract models of a system. Each model presents a different view or perspective of that system. Represent the system using some kind of graphical notation (Mostly based on UML). System modelling helps the analyst to understand the functionality of the system and models are used to communicate with customers. It also provides the developer and the customer with the means to assess quality, once the software is built.

Different Models different perspectives

You may develop different models to represent the system from different perspectives. For example:

- An **external perspective**, where you model the environment of the system.
- An **interaction perspective** where you model the interactions between a system and its environment or between the components of a system.
- A **structural perspective**, where you model the organization of a system or the structure of the data that is processed by the system.
- A **behavioural perspective**, where you model the dynamic behaviour of the system and how it responds to events

What is UML?

UML stands for Unified Modeling Language,

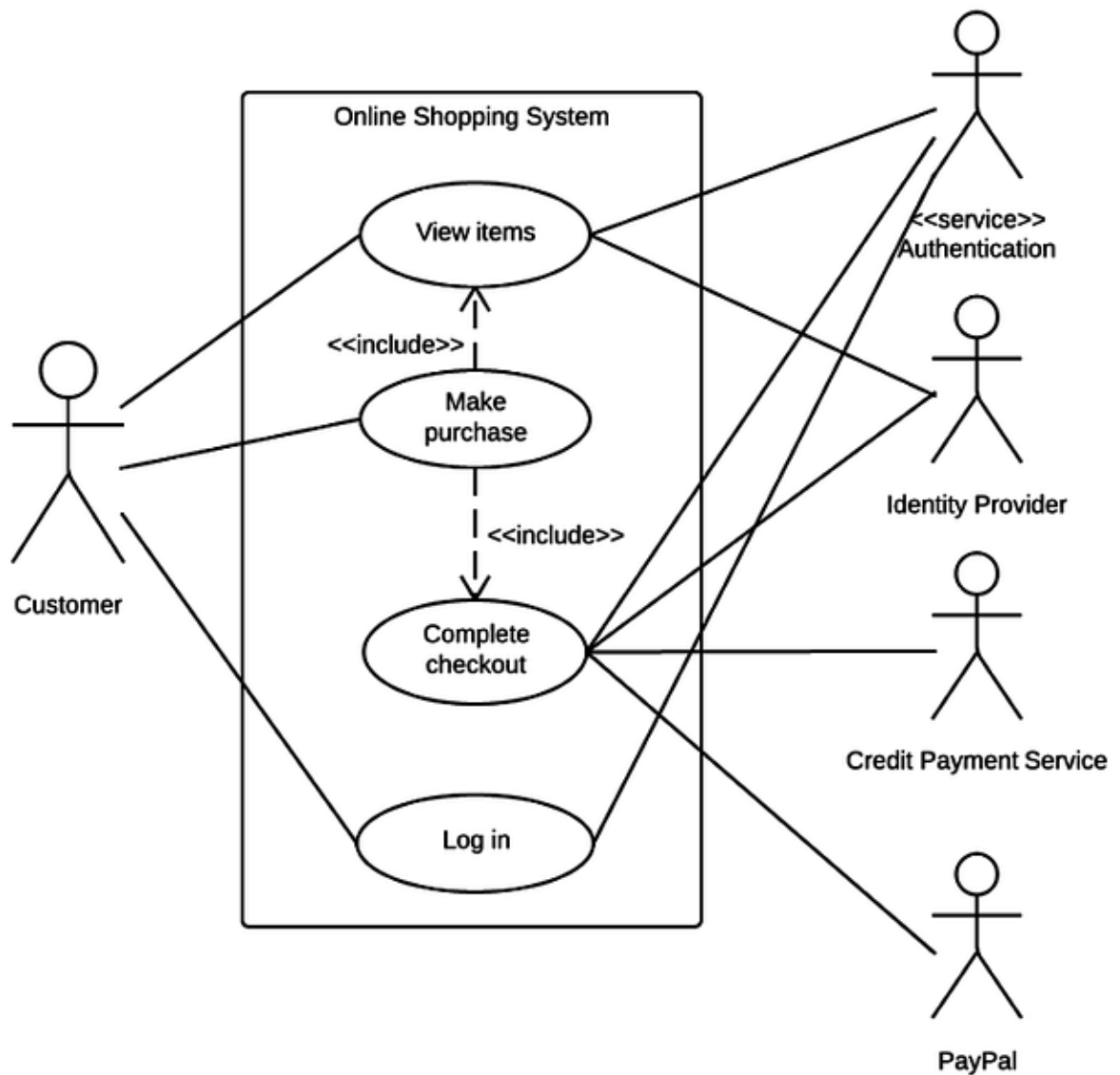
The Unified Modeling Language (UML) is a general-purpose, developmental, modelling language in the field of software engineering, that is intended to provide a standard way to visualize the design of a system.

UML diagrams represent two different views of a system model.

1. Static (or structural) view: emphasizes the static structure of the system using objects, attributes, operations and relationships. It includes class diagrams and composite structure diagrams.
2. Dynamic (or behavioural) view: emphasizes the dynamic behaviour of the system by showing collaborations among objects and changes to the internal states of objects. This view includes sequence diagrams, activity diagrams and use case diagrams.

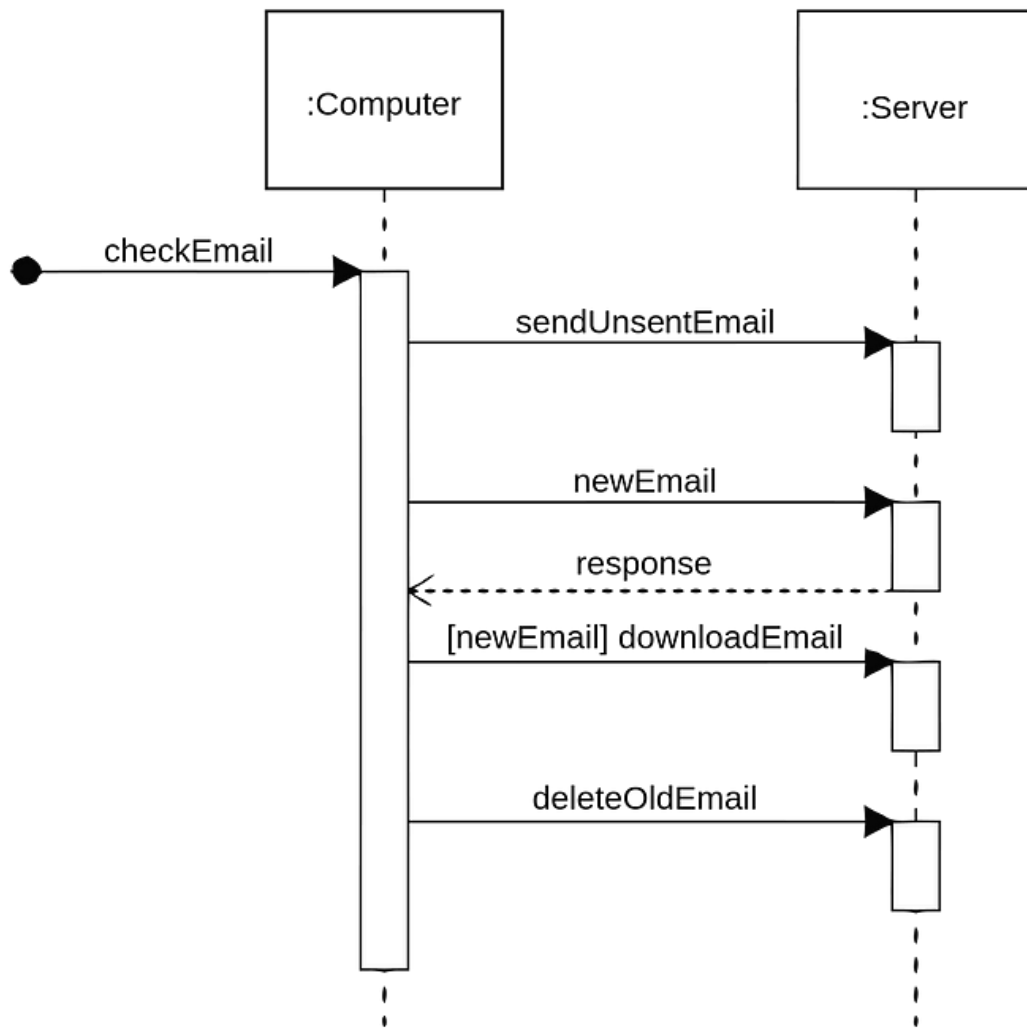
Essential UML diagrams for system modelling

Use **case diagrams**, which show the interactions between a system and its environment.

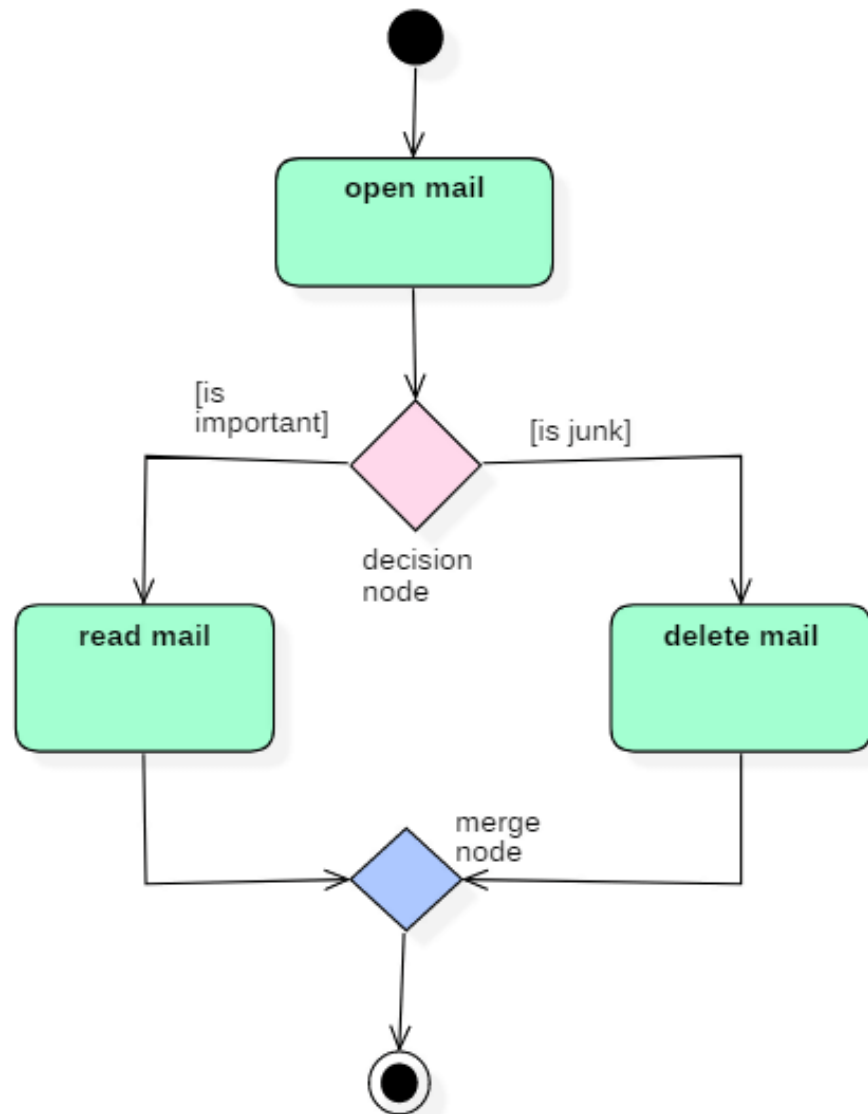


Online Shopping System Use Case diagram

Sequence diagrams, which show interactions between actors and the system and between system components.



Activity diagrams, which show the activities involved in a process or in the data processing.



Class diagrams, which show the object classes in the system and the associations between these classes.

